PROBABILITY DISTRIBUTIONS — QUICK REFERENCE

▣ DISCRETE DISTRIBUTIONS

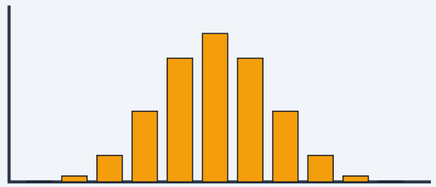
BINOMIAL — $\mathrm{Bin}(n,\,p)$

n = trials · p = P(success)

PMF $P(X{=}k) = \binom{n}{k} p^k (1-p)^{n-k}$

CDF $F(k) = I_{1-p}(n-k,\,k+1)$

E[X] np **VAR** $np(1-p)$



- Symmetric when $p = 0.5$
- Right-skew $p < 0.5$; left-skew $p > 0.5$
- $\mathrm{Bin} \to \mathcal{N}$ via CLT for large n
- $\mathrm{Bin}(n,p) \to \mathrm{Pois}(\lambda)$ if $n \to \infty,\ p \to 0$

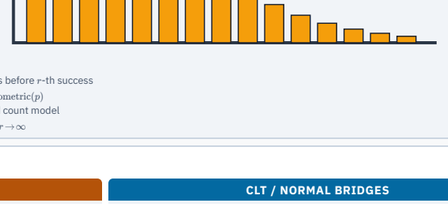
HYPERGEOMETRIC — $\mathrm{HG}(N,\,K,\,n)$

N = pop · K = successes in pop · n = draws

PMF $P(X{=}k) = \dfrac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$

CDF $F(k) = \sum_{i=0}^{k} \dfrac{\binom{K}{i}\binom{N-K}{n-i}}{\binom{N}{n}}$

E[X] $\dfrac{nK}{N}$ **VAR** $n\dfrac{K}{N}\dfrac{N-K}{N}$



- Sampling *without* replacement
- HG $\to$ Bin when $N \gg n$
- Finite population correction factor
- Used in acceptance sampling (MIL-STD)

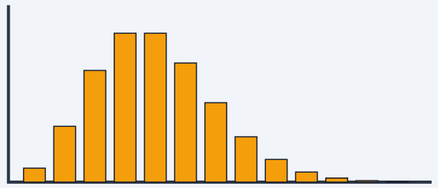
POISSON — $\mathrm{Pois}(\lambda)$

λ = average rate (events per interval)

PMF $P(X{=}k) = \dfrac{e^{-\lambda}\,\lambda^k}{k!}$

CDF $F(k) = e^{-\lambda} \sum_{j=0}^{k} \dfrac{\lambda^j}{j!} = \dfrac{\Gamma(k+1,\,\lambda)}{k!}$

E[X] λ **VAR** λ (mean = var)



- $\mathrm{E}[X] = \mathrm{Var}(X) = \lambda$ always
- Sum of Poissons is Poisson
- Interarrival times $\sim \mathrm{Exp}(\lambda)$
- $\to \mathcal{N}(\lambda,\lambda)$ for large λ

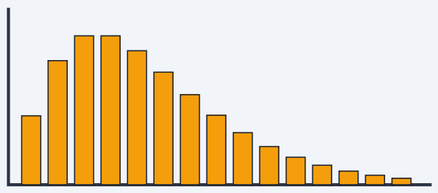
NEGATIVE BINOMIAL — $\mathrm{NB}(r,\,p)$

r = # successes · p = P(success) per trial

PMF $P(X{=}k) = \binom{k+r-1}{k}(1-p)^k p^r$

CDF $F(k) = I_p(r,\,k+1)$ (reg. incomplete beta)

E[X] $\dfrac{r(1-p)}{p}$ **VAR** $\dfrac{r(1-p)}{p^2}$



- Counts failures before r-th success
- $\mathrm{NB}(1,p) = \mathrm{Geometric}(p)$
- Overdispersed count model
- $\mathrm{NB} \to \mathrm{Pois}$ as $r \to \infty$

◉ CONTINUOUS DISTRIBUTIONS

NORMAL — $\mathcal{N}(\mu,\,\sigma^2)$

μ = mean · σ² = variance

PDF $f(x) = \dfrac{1}{\sigma\sqrt{2\pi}} \exp\left(-\dfrac{(x-\mu)^2}{2\sigma^2}\right)$

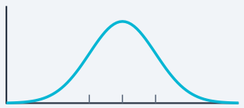
CDF $F(x) = \Phi\left(\dfrac{x-\mu}{\sigma}\right) = \dfrac{1}{2}\left[1 + \mathrm{erf}\left(\dfrac{x-\mu}{\sigma\sqrt{2}}\right)\right]$

E[X] μ **VAR** σ^2

z $Z = \dfrac{X-\mu}{\sigma} \sim \mathcal{N}(0,1)$

x̄ $Z = \dfrac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$ (σ known)

CI $\bar{X} \pm z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}}$



- 68–95–99.7 rule (1σ–2σ–3σ)
- Linear combos of normals are normal
- CLT: $\bar{X} \to \mathcal{N}(\mu, \sigma^2/n)$
- Basis of Z, t, χ^2, F statistics

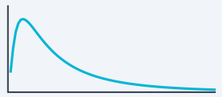
LOGNORMAL — $\mathrm{LN}(\mu,\,\sigma^2)$

μ, σ² = parameters of ln(X) ~ N(μ, σ²)

PDF $f(x) = \dfrac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\dfrac{(\ln x - \mu)^2}{2\sigma^2}\right)$

CDF $F(x) = \Phi\left(\dfrac{\ln x - \mu}{\sigma}\right)$

E[X] $e^{\mu+\sigma^2/2}$ **VAR** $(e^{\sigma^2}-1)\,e^{2\mu+\sigma^2}$



- $X = e^Y$ where $Y \sim \mathcal{N}(\mu,\sigma^2)$
- Right-skewed; support $x > 0$
- Mode $= e^{\mu-\sigma^2}$; Median $= e^{\mu}$
- Product of lognormals is lognormal

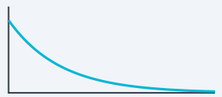
EXPONENTIAL — $\mathrm{Exp}(\lambda)$

λ = rate · mean = 1/λ

PDF $f(x) = \lambda\, e^{-\lambda x}, \quad x \geq 0$

CDF $F(x) = 1 - e^{-\lambda x}$

E[X] $\dfrac{1}{\lambda}$ **VAR** $\dfrac{1}{\lambda^2}$



- Memoryless: $P(X > s+t \mid X > s) = P(X > t)$
- Constant hazard rate $h(t) = \lambda$
- $=$ Weibull($\beta = 1,\ \eta = 1/\lambda$)
- CV = 1 always; interarrival of Poisson

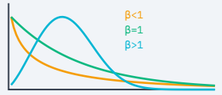
WEIBULL — $\mathcal{W}(\beta,\,\eta)$

β = shape · η = scale (char. life)

PDF $f(x) = \dfrac{\beta}{\eta}\left(\dfrac{x}{\eta}\right)^{\beta-1} \exp\left[-\left(\dfrac{x}{\eta}\right)^{\beta}\right]$

CDF $F(x) = 1 - \exp\left[-\left(\dfrac{x}{\eta}\right)^{\beta}\right]$

E[X] $\eta\,\Gamma\left(1+\frac{1}{\beta}\right)$ **VAR** $\eta^2\left[\Gamma\left(1+\frac{2}{\beta}\right) - \Gamma\left(1+\frac{1}{\beta}\right)^2\right]$



- $\beta < 1$: DFR — infant mortality
- $\beta = 1$: CFR — Exponential
- $\beta > 1$: IFR — wear-out failures
- $\beta \approx 3.44$: $\approx \mathcal{N}$; $\beta = 2$: Rayleigh

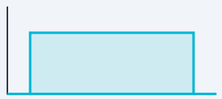
UNIFORM — $\mathcal{U}(a,\,b)$

a = lower bound · b = upper bound

PDF $f(x) = \dfrac{1}{b-a}, \quad a \leq x \leq b$

CDF $F(x) = \dfrac{x-a}{b-a}$

E[X] $\dfrac{a+b}{2}$ **VAR** $\dfrac{(b-a)^2}{12}$



- Maximum entropy given bounded support
- Basis of simulation (pseudo-RNG)
- $\mathcal{U}(0,1) \to$ any dist. via inverse CDF
- $\mathrm{Beta}(1,1) = \mathcal{U}(0,1)$

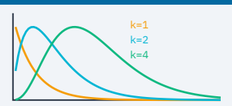
GAMMA — $\Gamma(k,\,\theta)$

k = shape · θ = scale (rate λ = 1/θ)

PDF $f(x) = \dfrac{x^{k-1}e^{-x/\theta}}{\theta^k\,\Gamma(k)}, \quad x > 0$

CDF $F(x) = \dfrac{\gamma(k,\,x/\theta)}{\Gamma(k)}$ (lower reg. inc. gamma)

E[X] $k\theta$ **VAR** $k\theta^2$



- $\Gamma(1,\lambda) = \mathrm{Exp}(\lambda)$
- $\Gamma(\nu/2,\,2) = \chi^2(\nu)$
- Sum of k Exp(λ) is $\Gamma(k,\,1/\lambda)$
- Conjugate prior for Poisson rate

△ SAMPLING / TEST DISTRIBUTIONS

STUDENT'S t — $t(\nu)$

ν = degrees of freedom

PDF $f(t) = \dfrac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1+\dfrac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$

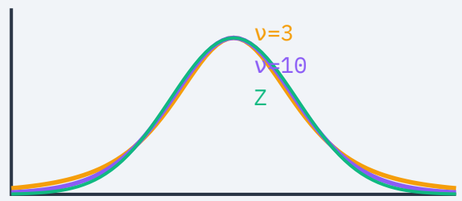
CDF $F(t) = I_{\frac{\nu}{\nu+t^2}}\left(\dfrac{\nu}{2},\,\dfrac{1}{2}\right)\cdot\dfrac{1}{2}$

E[X] $0\ (\nu > 1)$ **VAR** $\dfrac{\nu}{\nu-2}\ (\nu > 2)$

T $t = \dfrac{\bar{X}-\mu_0}{s/\sqrt{n}}, \quad \nu = n-1$

T₂ $t = \dfrac{X_1 - X_2}{s_p\sqrt{1/n_1 + 1/n_2}},\ \nu = n_1 + n_2 - 2$

CI $\bar{X} \pm t_{\alpha/2,\,\nu}\dfrac{s}{\sqrt{n}}$



- $t = Z/\sqrt{\chi^2/\nu}$
- $t \to Z(0,1)$ as $\nu \to \infty$
- Heavier tails than normal
- $t^2(\nu) = F(1,\nu)$

CHI-SQUARE — $\chi^2(\nu)$

ν = degrees of freedom

PDF $f(x) = \dfrac{x^{\nu/2-1}e^{-x/2}}{2^{\nu/2}\,\Gamma(\nu/2)}, \quad x \geq 0$

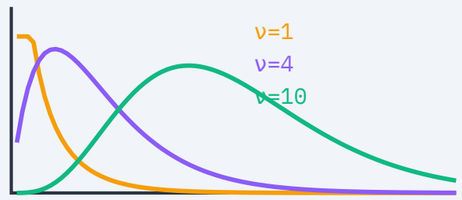
CDF $F(x) = \dfrac{\gamma(\nu/2,\ x/2)}{\Gamma(\nu/2)}$

E[X] ν **VAR** 2ν

X² $\chi^2 = \dfrac{(n-1)s^2}{\sigma_0^2}, \quad \nu = n-1$

GOF $\chi^2 = \sum_{i=1}^{k} \dfrac{(O_i - E_i)^2}{E_i},\ \nu = k-1$

IND $\chi^2 = \sum \dfrac{(O_{ij} - E_{ij})^2}{E_{ij}},\ \nu = (r-1)(c-1)$



- $\chi^2(\nu) = \sum_{i=1}^{\nu} Z_i^2$
- $= \Gamma(\nu/2,\,2)$
- $\to \mathcal{N}$ for large ν
- Used: variance tests, GOF, independence

F-DISTRIBUTION — $F(d_1,\,d_2)$

d_1 = numerator df · d_2 = denominator df

PDF $f(x) = \dfrac{\sqrt{\frac{(d_1x)^{d_1}\,d_2^{d_2}}{(d_1x+d_2)^{d_1+d_2}}}}{x\,B\left(\frac{d_1}{2},\,\frac{d_2}{2}\right)}$

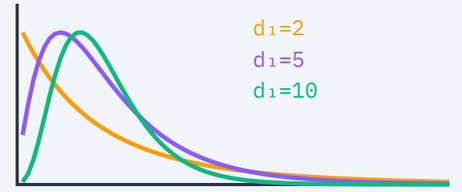
CDF $F(x) = I_{\frac{d_1x}{d_1x+d_2}}\left(\dfrac{d_1}{2},\,\dfrac{d_2}{2}\right)$

E[X] $\dfrac{d_2}{d_2-2}$ **VAR** $\dfrac{2d_2^2(d_1+d}{d_1(d_2-2)^2(}$

F $F = \dfrac{s_1^2}{s_2^2}\ \ (\sigma_1^2 = \sigma_2^2)$

ANV $F = \dfrac{\mathrm{MS}_B}{\mathrm{MS}_W} = \dfrac{\mathrm{SS}_B/(k-1)}{\mathrm{SS}_W/(N-k)}$

REG $F = \dfrac{\mathrm{MS}_R}{\mathrm{MS}_e} = \dfrac{\mathrm{SS}_R/p}{\mathrm{SS}_e/(n-p-1)}$



- $F = \frac{\chi^2_{d_1}/d_1}{\chi^2_{d_2}/d_2}$
- $1/F(d_1,d_2) = F(d_2,d_1)$
- $F(1,\nu) = t^2(\nu)$
- ANOVA ratio: $F = \mathrm{MS}_B/\mathrm{MS}_W$

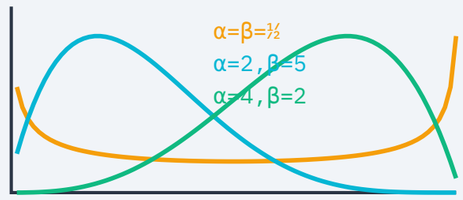
BETA — $\mathrm{Beta}(\alpha,\,\beta)$

α, β = shape parameters (both > 0)

PDF $f(x) = \dfrac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}, \quad 0 \leq x \leq 1$

CDF $F(x) = I_x(\alpha,\beta)$ (regularized incomplete beta)

E[X] $\dfrac{\alpha}{\alpha+\beta}$ **VAR** $\dfrac{\alpha\beta}{(\alpha+\beta)^2(\alpha+}$



- Support $[0,1]$; models proportions
- $\mathrm{Beta}(1,1) = \mathcal{U}(0,1)$
- $\alpha, \beta < 1 \Rightarrow$ U-shaped (bimodal)
- Conjugate prior for Binomial p

INTER-DISTRIBUTION RELATIONSHIPS & LIMITING CASES

DISCRETE LIMITS

- $\mathrm{HG}(N,K,n) \to \mathrm{Bin}(n, K/N)$ if $N \gg n$
- $\mathrm{Bin}(n,p) \to \mathrm{Pois}(\lambda)$ if $n \to \infty,\ p \to 0,\ np = \lambda$
- $\mathrm{NB}(r,p) \to \mathrm{Pois}$ as $r \to \infty$
- $\mathrm{Geom}(p) = \mathrm{NB}(1,p)$

CLT / NORMAL BRIDGES

- $\mathrm{Bin}(n,p) \to \mathcal{N}(np,\,np(1-p))$ large n
- $\mathrm{Pois}(\lambda) \to \mathcal{N}(\lambda,\lambda)$ large λ
- $\bar{X} \to \mathcal{N}(\mu, \sigma^2/n)$ by CLT
- Sum of any iid RVs $\to \mathcal{N}$ (CLT)

NORMAL → TEST STATS

- $Z^2 \sim \chi^2(1)\ ;\ \sum_i Z_i^2 \sim \chi^2(n)$
- $t(\nu) = Z/\sqrt{\chi^2(\nu)/\nu}$
- $F(d_1,d_2) = \frac{\chi^2_{d_1}/d_1}{\chi^2_{d_2}/d_2}$
- $t^2(\nu) = F(1,\nu)$

TEST STAT LIMITS

- $t(\nu) \to Z(0,1)$ as $\nu \to \infty$
- $\chi^2(\nu)/\nu \to 1$ as $\nu \to \infty$
- $\sqrt{2\chi^2(\nu)} - \sqrt{2\nu-1} \approx Z$
- $F(d_1,d_2) \to \chi^2(d_1)/d_1$ as $d_2 \to \infty$

CONTINUOUS FAMILIES

- $\mathrm{Exp}(\lambda) = \Gamma(1,1/\lambda) = \mathcal{W}(\beta{=}1,\eta{=}1/\lambda)$
- $\Gamma(\nu/2,\,2) = \chi^2(\nu)$
- $Y \sim \mathcal{N} \Rightarrow e^Y \sim \mathrm{LN}$
- $\mathrm{Beta}(1,1) = \mathcal{U}(0,1)$

POISSON ↔ EXPONENTIAL

- Events in $[0,t]$: count $\sim \mathrm{Pois}(\lambda t)$
- Interarrival time $T \sim \mathrm{Exp}(\lambda)$
- k -th event time $\sim \Gamma(k, 1/\lambda)$
- $P(N(t) \geq k) = P(T_k \leq t)$ duality

β GUIDE: β < 1→DFR(infant) β = 1→CFR=Exp β > 1→IFR(wear-out) β ≈ 3.44→≈ $\mathcal{N}$ β = 2→Rayleigh

$C_p = \frac{\mathrm{USL}-\mathrm{LSL}}{6\sigma}$ · $C_{pk} = \min\left[\frac{\mu-\mathrm{LSL}}{3\sigma}, \frac{\mathrm{USL}-\mu}{3\sigma}\right]$ · $Z = \frac{X-\mu}{\sigma}$ · $\chi^2 = \sum \frac{(O-E)^2}{E}$